

Ushtrime ND6b

1.3.73 Trupi me masë $m=2 \text{ kg}$ bënë lëvizje të njëtrajtëshme rrethore nëpër rrethin me rreze $r=0.5 \text{ m}$ duke i bërë 60 rrotullime në minutë. Të gjendet:

- Perioda?
- Frekuenca?
- Shpejtësia këndore?
- Shpejtësia lineare?
- Nxitimi radial?
- Forca centripetale?

Të dhënat:

$$m = 2 \text{ kg}$$

$$r = 0.5 \text{ m}$$

$$n = 60$$

$$t = 1 \text{ min} = 60 \text{ s}$$

a)

$$T = \frac{2\pi}{\omega}; T = \frac{1}{f}$$

$$a) T = \frac{t}{n} = \frac{60 \text{ s}}{60} = 1 \text{ s}$$

$$b) f = \frac{1}{T} = 1 \text{ Hz}$$

$$c) \varphi = \frac{l}{r} \Rightarrow \boxed{\varphi = \omega t}$$

$$r = \frac{v}{\omega} \Rightarrow \omega = \frac{v}{r}$$

$$\varphi = \frac{l}{\frac{v}{\omega}} = \frac{l \cdot \omega}{v} = \frac{l}{\frac{v}{\omega}} = \omega$$

$$\varphi = \omega \cdot t \Rightarrow \omega = \frac{\varphi}{t} = \frac{60 \cdot 2\pi \text{ rad}}{60 \text{ s}} = 2\pi \frac{\text{rad}}{\text{s}} = 6.28 \frac{\text{rad}}{\text{s}}$$

$$T = \frac{2\pi}{\omega} \Rightarrow \omega = \frac{2\pi}{T} = \frac{2\pi}{1 \text{ s}} = 2\pi \frac{\text{rad}}{\text{s}} = 6.28 \frac{\text{rad}}{\text{s}}$$

$$c) v = \omega \cdot r = 6.28 \frac{1}{\text{s}} \cdot 0.5 \text{ m} = 3.14 \frac{\text{m}}{\text{s}}$$

$$d) a_r = \frac{v^2}{r} = \frac{\left(3.14 \frac{\text{m}}{\text{s}}\right)^2}{0.5 \text{ m}} = \frac{9.8596 \frac{\text{m}^2}{\text{s}^2}}{0.5 \text{ m}} = 19.72 \frac{\text{m}}{\text{s}^2}$$

$$e) F_{cp} = \frac{mv^2}{r} = ma_r = 2 \text{ kg} \cdot 19.72 \frac{\text{m}}{\text{s}^2} = 39.44 \text{ N}$$

$$\boxed{F_{cp} = 39.44 \text{ N}} \quad \text{u}$$